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MODULE I

PARTIAL DIFFERENTIAL EQUATIONS

- * Partial differential Equations
- * Formation of partial differential equations - elimination of arbitrary constants - elimination of arbitrary functions.
- * Solutions of a partial differential equations.
- * Equations solvable by direct integration
- * Linear equations of the first order - Lagrange's linear equation
- * Non-linear equations of first order - Charpit's method.
- * Solution of equation by method of separation of variables.

Partial Differential Equations.

Partial differential equations arise by the application of the laws of physics in engineering and other fields when the number of independent variables is two or more. An equation in which one or more partial derivatives of an unknown function of two or more independent variables occurs is called a partial differential equation (pde). The order of a pde is the order of the highest derivatives which occurs in the equation and the degree of the equation is the degree of the highest derivative.

Examples:

i) $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 3z$ is a pde of first order and first degree.

ii) $\frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = x^2 + y^2$, 2nd order, 1st degree

iii) $\left(\frac{\partial^2 z}{\partial x^2}\right)^3 + \frac{\partial z}{\partial y} = 0$, 2nd order, 3rd degree.

If the dependent variable and its partial derivatives occurs linearly, we say that pde is linear.

We consider pde's of orders one and two in two independent variables only since these are the ones which mostly occur in physical applications. If z is a function of x and y , then we shall use the following notations.

$$p = \frac{\partial z}{\partial x}, \quad q = \frac{\partial z}{\partial y}, \quad r = \frac{\partial^2 z}{\partial x^2}, \quad s = \frac{\partial^2 z}{\partial x \partial y}, \quad t = \frac{\partial^2 z}{\partial y^2}.$$

The most general form of first order pde is $F(x, y, z, p, q) = 0$ and that of 2nd order linear pde is

$$a_1 \frac{\partial^2 z}{\partial x^2} + a_2 \frac{\partial^2 z}{\partial x \partial y} + a_3 \frac{\partial^2 z}{\partial y^2} + a_4 \frac{\partial z}{\partial x} + a_5 \frac{\partial z}{\partial y} + a_6 z = a_7$$

where a_i 's are the functions of x and y .

Formation of Partial Differential Equations.

A pde is formed by two methods

- i) by the elimination of arbitrary constants
- ii) by the elimination of arbitrary functions.

Elimination of Arbitrary constants.

If the number of arbitrary constants to be eliminated is equal to the number of independent variables the method of elimination

will give rise to pde of first-order. If the number of arbitrary constants to be eliminated is more than the number of independent variables, the process of elimination will give rise to pde of second and higher orders.

Problems:

Form pde from the following equations by eliminating the arbitrary constants.

i) $z = a^2x + b^2y + ab$

iv) $z = xy + y\sqrt{x^2 + y^2} + b$

ii) $(x-a)^2 + (y-b)^2 + z^2 = c^2$

v) $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

iii) $(x-a)^2 + (y-b)^2 = z^2 \cot^2 \alpha$

vi) $A(1+a^2)z = (x+ay+b)^2$

Solutions

i) $z = a^2x + b^2y + ab \rightarrow \textcircled{1}$

Differentiate $\textcircled{1}$ partially w.r.t x and y .

$$\frac{\partial z}{\partial x} = a^2 \Rightarrow p = a^2 \rightarrow \textcircled{2}, \quad \frac{\partial z}{\partial y} = b^2 \Rightarrow q = b^2 \rightarrow \textcircled{3}$$

Eliminating a and b using $\textcircled{1}$, $\textcircled{2}$ & $\textcircled{3}$ we get.

$$z = \underline{px + qy + pq}$$

ii) $(x-a)^2 + (y-b)^2 + z^2 = c^2 \rightarrow \textcircled{1}$

Differentiate $\textcircled{1}$ partially w.r.t x & y .

$$2(x-a) + 2z \cdot \frac{\partial z}{\partial x} = 0 \Rightarrow (x-a) = -zp \rightarrow \textcircled{2}$$

$$2(y-b) + 2z \cdot \frac{\partial z}{\partial y} = 0 \Rightarrow (y-b) = -zq \rightarrow \textcircled{3}$$

Eliminating a and b using ①, ② & ③, we get:

$$\textcircled{1} \Rightarrow (-zp)^2 + (-zq)^2 + z^2 = c^2 \Rightarrow z^2(\underline{p^2 + q^2 + 1}) = c^2.$$

ii) $(x-a)^2 + (y-b)^2 = z^2 \cot^2 \alpha. \quad \text{--- ①}$

Differentiating ① partially w.r.t x & y .

$$2(x-a) = 2zp \cot^2 \alpha \Rightarrow x-a = zp \cot^2 \alpha \quad \text{--- ②}$$

$$2(y-b) = 2zq \cot^2 \alpha \Rightarrow y-b = zq \cot^2 \alpha \quad \text{--- ③}$$

Eliminating a and b using ①, ② and ③

$$\textcircled{1} \Rightarrow (zp \cot^2 \alpha)^2 + (zq \cot^2 \alpha)^2 = z^2 \cot^2 \alpha.$$

$$\Rightarrow z^2 \cot^4 \alpha (p^2 + q^2) = z^2 \cot^2 \alpha$$

$$\underline{p^2 + q^2 = \tan^2 \alpha.}$$

iv) $z = xy + y\sqrt{x^2 + a^2} + b. \quad \text{--- ①}$

Differentiating ① partially w.r.t x & y

$$\frac{\partial z}{\partial x} = y + y \cdot \frac{1}{2\sqrt{x^2 + a^2}} \cdot 2x \Rightarrow p = y + \frac{xy}{\sqrt{x^2 + a^2}} \quad \text{--- ②}$$

$$\frac{\partial z}{\partial y} = x + \sqrt{x^2 + a^2} \Rightarrow q = x + \sqrt{x^2 + a^2} \quad \text{--- ③}$$

$$\Rightarrow q - x = \sqrt{x^2 + a^2}$$

Eliminating a and b using ①, ② & ③

$$\textcircled{2} \Rightarrow p - y = \frac{xy}{q - x} \Rightarrow (p - y)(q - x) = xy.$$

$$\Rightarrow pq - px - yq + xy = xy \Rightarrow \underline{px + qy = pq.}$$

$$v) z = \frac{x^2}{a^2} + \frac{y^2}{b^2} \quad \text{--- (1)}$$

Differentiating (1) partially w.r.t x & y

$$2 \frac{\partial z}{\partial x} = \frac{2x}{a^2} \Rightarrow p = \frac{x}{a^2} \Rightarrow a^2 = \frac{x}{p} \quad \text{--- (2)}$$

$$2 \frac{\partial z}{\partial y} = \frac{2y}{b^2} \Rightarrow q = \frac{y}{b^2} \Rightarrow b^2 = \frac{y}{q} \quad \text{--- (3)}$$

From (1), (2) & (3)

$$z = \frac{x^2}{(x/p)} + \frac{y^2}{(y/q)} \Rightarrow z = \underline{\underline{px + qy}}$$

$$vi) z = (1 + a^2)^{\frac{1}{2}} = (x + ay + b)^{\frac{1}{2}} \quad \text{--- (1)}$$

Differentiating (1) partially w.r.t x & y

$$a(1+a^2)^{\frac{1}{2}} \frac{\partial z}{\partial x} = \frac{1}{2}(x+ay+b)^{-\frac{1}{2}} \Rightarrow a(1+a^2)^{\frac{1}{2}} p = \frac{1}{2}(x+ay+b)^{-\frac{1}{2}} \quad \text{--- (2)}$$

$$a(1+a^2)^{\frac{1}{2}} \frac{\partial z}{\partial y} = \frac{1}{2}(x+ay+b)^{-\frac{1}{2}} a \Rightarrow a(1+a^2)^{\frac{1}{2}} q = \frac{1}{2}a(x+ay+b)^{-\frac{1}{2}} \quad \text{--- (3)}$$

From (2) & (3) $\frac{p}{q} = \frac{1}{a}$

$$(2) \Rightarrow x + ay + b = a^2 \left(1 + \frac{q^2}{p^2}\right) p$$

$$\begin{aligned} (1) \Rightarrow z &= \left(1 + \frac{q^2}{p^2}\right)^{\frac{1}{2}} = \left(1 + \frac{q^2}{p^2}\right)^{\frac{1}{2}} \cdot p \\ &\Rightarrow \underline{\underline{z = p^2 + q^2}} \end{aligned}$$

Elimination of Arbitrary functions.

The elimination of one arbitrary function from a given relation leads to a partial differential equation of order one. But the elimination of two arbitrary functions leads to a pde of order two.

Problems:

Form the pde by eliminating the arbitrary functions from the following equations.

i) $z = (x+y) f(x^2-y^2)$

iv) $z = f(x+at) + g(x-at)$

ii) $z = y^2 + 2f\left(\frac{1}{x} + \log y\right)$

v) $z = f(2x+y) + g(3x-y)$

iii) $z = x f\left(\frac{y}{x}\right) + y \cdot g(x)$

vi) $z = x f(2x-y) + g(2x-y)$

Solution

i) $z = (x+y) f(x^2-y^2) \quad \text{--- (1)}$

Differentiating (1) partially w.r.t x and y .

$$\frac{\partial z}{\partial x} = (x+y) f'(x^2-y^2) 2x + f(x^2-y^2)$$

$$\Rightarrow p = 2x(x+y) f'(x^2-y^2) + \frac{z}{x+y}$$

$$p - \frac{z}{x+y} = 2x(x+y) f'(x^2-y^2) \quad \text{--- (2)}$$

$$\frac{\partial z}{\partial y} = (x+y) f'(x^2-y^2) (-2y) + f(x^2-y^2)$$

$$\Rightarrow q = -2y(x+y) f'(x^2-y^2) + \frac{z}{x+y}$$

$$q - \frac{z}{x+y} = -2y(x+y) f'(x^2-y^2) \quad \text{--- (3)}$$

$$\frac{(2)}{(3)} \Rightarrow \frac{p - \frac{z}{x+y}}{q - \frac{z}{x+y}} = -\frac{x}{y}$$

$$pxy + py^2 - zy + x^2q + xyq - xz = 0$$

$$py(x+y) + qx(x+y) - z(x+y) = 0$$

$$\underline{py + qx = z}$$

ii) $z = y^2 + 2f\left(\frac{1}{x} + \log y\right)$ — (1)

Differentiating (1) partially w.r.t x & y

$$p = \frac{\partial z}{\partial x} = 2f'\left(\frac{1}{x} + \log y\right) \cdot \left(-\frac{1}{x^2}\right)$$
 — (2)

$$q = \frac{\partial z}{\partial y} = 2y + 2f'\left(\frac{1}{x} + \log y\right) \cdot \frac{1}{y}$$

$$\Rightarrow q - 2y = 2f'\left(\frac{1}{x} + \log y\right) \cdot \frac{1}{y}$$
 — (3)

$$\frac{(2)}{(3)} \Rightarrow \frac{p}{q - 2y} = \frac{-\frac{y}{x^2}}{\frac{1}{y}} = -\frac{y}{x^2}$$

$$\Rightarrow x^2 p = -yq + 2y^2 \Rightarrow \underline{2y^2 + yq = x^2 p}$$

iii) $z = x f(y/x) + y g(x)$ — (1)

Differentiating (1) partially w.r.t x & y

$$p = \frac{\partial z}{\partial x} = x f'(y/x) \cdot \left(-\frac{y}{x^2}\right) + f(y/x) + y g'(x)$$

$$= -\frac{y}{x} f'(y/x) + f(y/x) + y g'(x)$$

$$x p = -y f'(y/x) + x f(y/x) + xy g'(x)$$
 — (2)

$$q = \frac{\partial z}{\partial y} = x \cdot f'(y/x) = \frac{1}{x} + g(x).$$

$$= f'(y/x) + g(x)$$

$$\Rightarrow yq = y f'(y/x) + y \cdot g(x). \text{--- (3)}$$

$$\text{Also } t = \frac{\partial^2 z}{\partial y^2} = f''(y/x) \cdot \frac{1}{x}$$

$$\Rightarrow xt = f''(y/x) \text{--- (4)}$$

$$s = \frac{\partial^2 z}{\partial x \partial y} = f''(y/x) \cdot (-y/x^2) + g'(x). \text{--- (5)}$$

Eliminating the functions from (1), (2), (3), (4) & (5)

$$\text{Consider } xp + yq = -y' f'(y/x) + x f'(y/x) + xy g'(x) \\ + y f'(y/x) + y g(x)$$

$$xp + yq = x f'(y/x) + xy g'(x) + y g(x).$$

$$xp + yq = z + xy g'(x) \text{--- (6)}$$

$$\text{From (5) } g'(x) = s + \frac{y}{x^2} \cdot f''(y/x)$$

$$= s + \frac{y}{x^2} \cdot xt = s + \frac{y}{x} t.$$

$$\therefore (6) \text{ becomes } xp + yq = z + xy \left[s + \frac{y}{x} t \right]$$

$$\underline{\underline{xp + yq = z + xys + y^2 t.}}$$

$$= f(x+at) + g(x-at) \text{--- (7)}$$

Differentiating (1) partially w.r.t x and t .

$$p = \frac{\partial z}{\partial x} = f'(x+at) + a f'(x-at)$$

$$\frac{\partial z}{\partial t} = a f'(x+at) - a g'(x-at)$$

$$\frac{\partial^2 z}{\partial x^2} = f''(x+at) + g''(x-at) \quad \text{--- (2)}$$

$$\frac{\partial^2 z}{\partial t^2} = a^2 f''(x+at) + a^2 g''(x-at) \quad \text{--- (3)}$$

from (2) & (3) $\frac{\partial^2 z}{\partial t^2} = a^2 \cdot \frac{\partial^2 z}{\partial x^2}$

v) $z = f(2x+y) + g(3x-y)$

$$p = \frac{\partial z}{\partial x} = 2f'(2x+y) + 3g'(3x-y)$$

$$q = \frac{\partial z}{\partial y} = f'(2x+y) - g'(3x-y)$$

$$r = \frac{\partial^2 z}{\partial x^2} = 2f''(2x+y) + 3g''(3x-y)$$

$$s = \frac{\partial^2 z}{\partial x \partial y} = 2f''(2x+y) - 3g''(3x-y)$$

$$t = \frac{\partial^2 z}{\partial y^2} = f''(2x+y) + g''(3x-y)$$

$$r + s = 6f''(2x+y) + 6g''(3x-y) = 6t$$

$$\underline{r + s = 6t}$$

vi) $z = x f(2x-y) + g(2x-y)$

$$p = \frac{\partial z}{\partial x} = x f'(2x-y) + f(2x-y) + 2g'(2x-y)$$

$$q = \frac{\partial z}{\partial y} = -x f'(2x-y) - g'(2x-y)$$

$$r = \frac{\partial^2 z}{\partial x^2} = f''(2x-y) + f'(2x-y) + 2f''(2x-y) + 4g''(2x-y)$$

$$s = \frac{\partial^2 z}{\partial x \partial y} = -2x f''(2x-y) - f'(2x-y) + 2g'(2x-y)$$

$$t = \frac{\partial^2 z}{\partial y^2} = x f''(2x-y) + g''(2x-y)$$

$$s+t = -x f''(2x-y) - f'(2x-y) - g''(2x-y).$$

$$4s+4t = -[4x f''(2x-y) + 4f'(2x-y) + 4g''(2x-y)]$$

$$= -\phi.$$

$$\Rightarrow \underline{\underline{4s+4t+\phi = 0}}$$

Elimination of arbitrary function ϕ from the relation $\phi(u,v)=0$ where u and v are two functions of x, y and z .

$$\text{Let } \phi(u,v) = 0$$

$$\begin{vmatrix} u_x + u_z p & v_x + v_z p \\ u_y + u_z q & v_y + v_z q \end{vmatrix} = 0$$

which on simplification gives a pde of the form $Pp + Qq = R$ where P, Q, R are functions of x, y and z .

Problems:

Eliminate the arbitrary function from the following equation.

i) $\phi(x^2+y^2+z^2, z^2-2xy) = 0$

ii) $\phi(x^2+y^2+z^2, xyz) = 0$

iii) $\phi(xyz - \frac{1}{x} - \frac{1}{y} - \frac{1}{z}) = 0$

$$1) \quad u = x^2 + y^2 + z^2, \quad v = z^3 - 2xyz$$

$$u_x = 2x, \quad u_y = 2y, \quad u_z = 2z$$

$$v_x = -2y, \quad v_y = -2x, \quad v_z = 3z^2$$

The required eliminant is

$$\begin{vmatrix} u_x + u_z p & v_x + v_z p \\ u_y + u_z q & v_y + v_z q \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 2x + 2zp & -2y + 3zp \\ 2y + 2zq & -2x + 3zq \end{vmatrix} = 0$$

$$\Rightarrow (2x + 2zp)(-2x + 3zq) - (-2y + 3zp)(2y + 2zq) = 0$$

$$\Rightarrow -x^2 + 2zq - xz p + z^2 q + y^2 + 2yzq - zyp - z^2 q = 0$$

$$\Rightarrow -pz(x+y) + qz(x+y) = x^2 - y^2$$

$$\Rightarrow -pz(x+y) + qz(x+y) = (x+y)(x-y)$$

$$\Rightarrow \underline{\underline{-pz + qz = x - y}}$$

$$2. \quad u = x^2 + y^2 + z^2, \quad v = xyz$$

$$u_x = 2x, \quad u_y = 2y, \quad u_z = 2z$$

$$v_x = yz, \quad v_y = xz, \quad v_z = xy$$

The required eliminant is

$$\begin{vmatrix} u_x + u_z p & v_x + v_z p \\ u_y + u_z q & v_y + v_z q \end{vmatrix} = 0$$

$$\begin{vmatrix} 2x + 2zp & yz + xy p \\ 2y + 2zq & xz + xy q \end{vmatrix} = 0$$

$$\begin{aligned}
 (2x + 2zp)(x^2 + xyp) - (y^2 + xyp)(2y + 2zq) &= 0 \\
 2x^3 + 2x^2yp + 2z^2px + 2xypzpq - 2y^3 - 2y^2q \\
 - 2xy^2p - 2xypzpq &= 0
 \end{aligned}$$

$$x(x^2 - y^2)p + y(x^2 - z^2)q = x(y^2 - x^2).$$

iii) $u = xy + z^2, \quad v = z - xy.$

$$u_x = y, \quad u_y = x, \quad u_z = 2z$$

$$v_x = -y, \quad v_y = -x, \quad v_z = 1$$

The required eliminant is

$$\begin{vmatrix} u_x + u_z p & u_x + v_z p \\ u_y + u_z q & v_y + v_z q \end{vmatrix} = 0$$

$$\begin{vmatrix} y + 2z p & x + z p \\ x + 2z q & -x + q \end{vmatrix} = 0$$

$$(y + 2zp)(-x + q) - (x + zp)(x + 2zq) = 0$$

$$\underline{(qy - px)(1 + 2z) = 0}$$

Solutions of Partial differential equations.

A pde has two types of solutions, the solution containing arbitrary constants and the solution containing arbitrary functions.

For example, $z = ax + by$ and $z = x f(x/y)$ are solutions of the same equation $z = px + qy$

A solution which contains the maximum possible number of arbitrary constants is called a complete integral or complete solution

Thus $z = ax + by$ is a complete integral of $z = px + qy$.

A solution which contains the maximum possible number of arbitrary functions is called a general integral or general solution

Thus $z = x f(x/y)$ is a general integral of $z = px + qy$.

A solution which is obtained by giving specific values to one or more of the arbitrary constants or arbitrary functions in the complete integral or general integral is called a particular solution or particular integral

Thus $z = 3x + 4y$ and $z = x \cos(x/y)$ are

particular integrals of the equation $z = px + qy$.

A solution of a pde having no arbitrary constants or arbitrary functions and which cannot be obtained from the complete integral or general integral by giving specific values to the arbitrary constants or arbitrary functions is called singular solution of the pde.

Equations solvable by direct integration.

In a pde if the dependent variable z occurs only in the form of partial derivatives, then the pde can be solved by direct integration.

Problems

1. Solve $\frac{\partial^2 z}{\partial x \partial y} = \sin x$

Soln Given $\frac{\partial^2 z}{\partial x \partial y} = \sin x$ or $\frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \sin x$.

Integrating both sides w.r.t y , we get

$$\frac{\partial z}{\partial x} = y \sin x + f(x).$$

Integrating this w.r.t x , we get

$$\begin{aligned} z &= -y \cos x + \int f(x) dx + g(y) \\ &= -y \cos x + F(x) + g(y) \quad F(x) = \int f(x) dx \end{aligned}$$

2. Solve $\frac{\partial^2 z}{\partial y^2} = \sin(xy)$

Soln Given $\frac{\partial^2 z}{\partial y^2} = \sin(xy)$

Integrating w.r.t y , $\frac{\partial z}{\partial y} = -\frac{\cos xy}{x} + f(x)$

Integrating w.r.t y , $z = -\frac{\sin xy}{x^2} + \int f(x) dy + g(x)$

$\therefore z = -\frac{\sin xy}{x^2} + y f(x) + g(x)$

where $f(x)$ and $g(x)$ are arbitrary functions.

3. Solve $\frac{\partial z}{\partial x} = 6x + 3y$; $\frac{\partial z}{\partial y} = 3x - 4y$.

Soln Given $\frac{\partial z}{\partial x} = 6x + 3y$ Integrating w.r.t x ,
 $z = 3x^2 + 3xy + f(y)$ ——— (1)

Also $\frac{\partial z}{\partial y} = 3x - 4y$ Integrating w.r.t y .
 $z = 3xy - 2y^2 + g(x)$ ——— (2)

Equating (1) & (2) we get $g(x) = 3x^2$, and
 $f(y) = -2y^2$.

$\therefore z = 3x^2 + 3xy - 2y^2 + c$, where c is a constant.

4. Solve $\frac{\partial^2 z}{\partial x^2} + z = 0$, given that when $x=0$,

$z = e^y$ and $\frac{\partial z}{\partial x} = 1$.

Sol If z is a function of x alone, then $z = A \cos x + B \sin x$ is the solution of $\frac{\partial^2 z}{\partial x^2} + z = 0$

Since z is a function of both x and y , A and B are functions of y . Hence the solution of the given equation is

$$z = f(y) \cos x + g(y) \sin x \quad \text{--- (1)}$$

When $x=0$, $z = e^y \quad \therefore (1) \Rightarrow e^y = f(y)$.

$$\therefore z = e^y \cos x + g(y) \sin x. \quad \text{--- (2)}$$

$$\frac{\partial z}{\partial x} = -e^y \sin x + g(y) \cos x. \quad \text{--- (2)}$$

We've $\frac{\partial z}{\partial x} = 1$ when $x=0 \quad \therefore (2) \Rightarrow 1 = g(y)$

$$\therefore z = \underline{e^y \cos x + \sin x}.$$

5. Solve $\frac{\partial^2 z}{\partial x^2} = a^2 z$ given that when $x=0$,

$$\frac{\partial z}{\partial x} = a \sin y \quad \text{and} \quad \frac{\partial z}{\partial y} = 0.$$

Sol If z is a function of x alone, then $z = A e^{ax} + B e^{-ax}$. Since z is a function of both x & y , A & B are functions of y .

$$\therefore z = f(y) e^{ax} + g(y) e^{-ax} \quad \text{--- (1)}$$

differentiating (1) partially w.r.t x and y .

$$\frac{\partial z}{\partial x} = a f(y) e^{ax} - a g(y) e^{-ax}, \quad \frac{\partial z}{\partial y} = f'(y) e^{ax} + g'(y) e^{-ax}.$$

We've $\frac{\partial z}{\partial x} = a \sin y$ when $x=0$

$$\therefore a \sin y = a [f(y) - g(y)]$$

$$\sin y = f(y) - g(y) \longrightarrow (2)$$

Also we've $\frac{\partial z}{\partial y} = 0$ when $x=0$

$$\therefore 0 = f'(y) + g'(y) \Rightarrow f(y) + g(y) = A \quad (3)$$

$$(2) + (3) \Rightarrow 2f(y) = \sin y + A \quad \text{or} \quad f(y) = \frac{\sin y + A}{2}$$

$$(3) - (2) \Rightarrow 2g(y) = A - \sin y \quad \text{or} \quad g(y) = \frac{A - \sin y}{2}$$

$$\therefore z = \left(\frac{\sin y + A}{2} \right) e^{ay} + \left(\frac{A - \sin y}{2} \right) e^{-ay}$$

6. Solve $\frac{\partial^2 z}{\partial x \partial y} = \sin x \sin y$ for which $\frac{\partial z}{\partial y} = -2 \sin y$ when $x=0$ and $z=0$, when y is an odd multiple of $\pi/2$.

Soln Given $\frac{\partial^2 z}{\partial x \partial y} = \sin x \sin y$

Integrating w.r.t x , we get

$$\frac{\partial z}{\partial y} = -\cos x \sin y + f(y) \quad (1)$$

$$\text{When } x=0, \quad -2 \sin y = -\sin y + f(y) \\ \Rightarrow f(y) = -\sin y$$

$$\therefore (1) \Rightarrow \frac{\partial z}{\partial x} = -\cos x \sin y - \sin y$$

Integrating w.r.t y

$$z = \cos x \cos y + \cos y + g(x).$$

When $y = (2n+1)\frac{\pi}{2}$, $z = 0$

$$\therefore 0 = g(x) \quad \text{Since } \cos(2n+1)\frac{\pi}{2} = 0.$$

$$\therefore z = \cos x \cos y + \cos y = \underline{\underline{\cos y (\cos x + 1)}}$$

Linear Equations of First Order.

A linear partial differential equation of the first order commonly known as Lagrange's linear equation is of the form $Pp + Qq = R$ where P, Q and R are functions of x, y, z .

Method.

Step I: Form the Lagrange's subsidiary equations $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

Step II: Solve the above simultaneous equations by method of grouping or multipliers method.

Step III: If $u=a$ and $v=b$ are two independent solutions of the subsidiary equations then $\phi(u,v)=0$ where ϕ is an arbitrary function, is the general integral of the h.c.

Method of Grouping:

This method is applicable only when one of the variables is absent from two terms in the subsidiary equation. Suppose z is absent from P and Q . Then $\frac{dx}{P} = \frac{dy}{Q}$ is an ordinary differential equation connecting x and y and it gives a solution of the form $u=a$.

Multiplier method.

Let the auxiliary equations be

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\text{Thus we've } \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{l dx + m dy + n dz}{lP + mQ + nR}$$

where l, m, n may be constants or functions of x, y and z , l, m, n are chosen in such a way that $lP + mQ + nR = 0$.

$$\text{Thus } l dx + m dy + n dz = 0$$

Solve this differential equation and we get the solution as $u=a$.

Problems.

Solve the following.

i) $\tan x p + \tan y q = \tan z$.

ii) $\frac{y^2 z}{x} p + xz q = y^2$

iii) $(mz - ny)p + (nx - lz)q = ly - mx$

$$iv) \quad xp + yq = z$$

$$v) \quad (x^2 - y^2 - z^2)p + 2xyq = 2xz$$

$$vi) \quad (x^2 - yz)p + (y^2 - xz)q = z^2 - xy$$

Solutions

$$i) \quad A.E \text{ is } \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$i.e. \quad \frac{dx}{\tan x} = \frac{dy}{\tan y} = \frac{dz}{\tan z}$$

$$\cot x dx = \cot y dy$$

$$\text{Take } \frac{dx}{\tan x} = \frac{dy}{\tan y} \Rightarrow \text{Integrating both sides}$$

we get

$$\log \sin x = \log \sin y + \log a$$

$$\Rightarrow \log \left(\frac{\sin x}{\sin y} \right) = \log a \Rightarrow \frac{\sin x}{\sin y} = a$$

$$\text{Take } \frac{dy}{\tan y} = \frac{dz}{\tan z} \Rightarrow \cot y dy = \cot z dz$$

$$\text{Integrating both sides } \log \sin y = \log \sin z + \log b$$

$$\Rightarrow \log \left(\frac{\sin y}{\sin z} \right) = \log b$$

$$\Rightarrow \frac{\sin y}{\sin z} = b$$

$\therefore$ The required general solution is

$$\Phi \left(\frac{\sin x}{\sin y}, \frac{\sin y}{\sin z} \right) = 0, \quad \Phi \text{ is arbitrary.}$$

$$ii) \quad A.E \text{ is } \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{yz} = \frac{dy}{xz} = \frac{dz}{xz}$$

$$ii \quad \frac{dx}{y^2} = \frac{dy}{x^2} = \frac{dz}{xy^2} \quad (11)$$

Taking $\frac{dx}{y^2} = \frac{dy}{x^2} \Rightarrow x^2 dx = y^2 dy$

Integrating $\Rightarrow \frac{x^3}{3} = \frac{y^3}{3} + \frac{a}{3} \Rightarrow x^3 - y^3 = a$.

Taking $\frac{dx}{y^2} = \frac{dz}{xy^2} \Rightarrow x dx = z dz$

Integrating $\Rightarrow \frac{x^2}{2} = \frac{z^2}{2} + \frac{b}{2} \Rightarrow x^2 - z^2 = b$.

$\therefore$ The required general solution is

$$\phi(x^3 - y^3, x^2 - z^2) = 0.$$

ii) AE is $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$ii \quad \frac{dx}{mz - ny} = \frac{dy}{nx - lz} = \frac{dz}{ly - mx}$$

Choose multipliers $l = x, m = y, n = z$ so that
 $lP + mQ + nR = x(mz - ny) + y(nx - lz) + z(ly - mx) = 0$

$$l dx + m dy + n dz = 0 \Rightarrow x dx + y dy + z dz = 0$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = \frac{a}{2}$$

$$\Rightarrow x^2 + y^2 + z^2 = a.$$

Choose multipliers $l = l, m = m, n = n$ so that

$$lP + mQ + nR = l(mz - ny) + m(nx - lz) + n(ly - mx) = 0$$

$$l dx + m dy + n dz = 0 \Rightarrow lx + my + nz = b.$$

$\therefore$ general solution is $\phi(x^2 + y^2 + z^2, lx + my + nz) = 0$

iv) AE is $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

ii $\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z}$

Take $\frac{dx}{x} = \frac{dz}{z} \Rightarrow \log x = \log z + \log a$
 $\Rightarrow \log(x/z) = \log a \Rightarrow x/z = a$

Take $\frac{dy}{y} = \frac{dz}{z} \Rightarrow \log y = \log z + \log b$
 $\Rightarrow \log(y/z) = \log b \Rightarrow y/z = b$

$\therefore$ General solution is $\phi(x/z, y/z) = 0$

v) AE is $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

ii $\frac{dx}{x^2-y^2-z^2} = \frac{dy}{2xy} = \frac{dz}{2xz} = k$

Take $\frac{dy}{2xy} = \frac{dz}{2xz} \Rightarrow \log y = \log z + \log a$
 $\Rightarrow \log(y/z) = \log a \Rightarrow y/z = a$

Also $k = \frac{x dx + y dy + z dz}{x(x^2-y^2-z^2) + (2xy)y + (2xz)z}$
 $= \frac{x dx + y dy + z dz}{x^3 + xy^2 + xz^2} = \frac{x dx + y dy + z dz}{x(x^2+y^2+z^2)}$

Take $\frac{dz}{2xz} = \frac{x dx + y dy + z dz}{x(x^2+y^2+z^2)}$

$\Rightarrow \frac{dz}{z} = \frac{2x dx + 2y dy + 2z dz}{x^2+y^2+z^2}$

Integrating, $\log z = \log(x^2+y^2+z^2) + \log b$
 $\Rightarrow x^2+y^2+z^2 = b$

$\therefore$ general solution is $\phi\left(\frac{y}{z}, \frac{x^2+y^2+z^2}{z}\right) = 0$

i) AE is $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

ii $\frac{dx}{x^2-yz} = \frac{dy}{y^2-zx} = \frac{dz}{z^2-xy} = k$

Take $k = \frac{dx - dy}{x^2-yz - y^2+zx} = \frac{dy - dz}{y^2-zx - z^2+xy}$

$\Rightarrow \frac{dx - dy}{(x^2-y^2)+z(x-y)} = \frac{dy - dz}{(y^2-z^2)+x(y-z)}$

$\Rightarrow \frac{dx - dy}{(x+y)(x-y)+z(x-y)} = \frac{dy - dz}{(y+z)(y-z)+x(y-z)}$

$\Rightarrow \frac{dx - dy}{(x-y)(x+y+z)} = \frac{dy - dz}{(y-z)(x+y+z)}$

$\Rightarrow \frac{dx - dy}{x - y} = \frac{dy - dz}{y - z}$

Integrating $\log(x-y) = \log(y-z) + \log a$
 $\Rightarrow \frac{x-y}{y-z} = a$

Take $k = \frac{dx + dy + dz}{x^2-yz + y^2-zx + z^2-xy} = \frac{x dx + y dy + z dz}{x^3 - xyz + y^3 - xyz + z^3 - xyz}$

$\Rightarrow \frac{dx + dy + dz}{x^3 + y^3 + z^3 - 3xyz} = \frac{x dx + y dy + z dz}{x^3 + y^3 + z^3 - 3xyz}$

$\frac{dx + dy + dz}{x^3 + y^3 + z^3 - 3xyz} = \frac{x dx + y dy + z dz}{(x+y+z)(x^2+xy+yz+zx)}$

$$\Rightarrow dx + dy + dz = \frac{x dx + y dy + z dz}{x + y + z}$$

$$\Rightarrow (x+y+z)(dx+dy+dz) = x dx + y dy + z dz$$

$$\text{Integrating} \Rightarrow \frac{(x+y+z)^2}{2} = \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} + b$$

$$\Rightarrow \frac{x^2 + y^2 + z^2 + 2xy + 2yz + 2xz}{2} = \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} + b$$

$$\Rightarrow xy + yz + xz = b.$$

$\therefore$ the general solution is

$$\& \left(\frac{x-y}{y-z}, \underline{xy + yz + xz} \right) = 0$$

Non-linear Equation of First-order
Charpit's Method.

Charpit's method is a general method for solving first order non-linear partial differential equations in two independent variables. We want to find the complete integral for a non-linear first order pde $F(x, y, z, p, q) = 0$

This pde is solved by using the subsidiary equations

$$\frac{dp}{F_x + F_z p} = \frac{dq}{F_y + F_z q} = \frac{dz}{pF_p - qF_q} = \frac{dx}{-F_p} = \frac{dy}{-F_q}$$

These equations are known as Charpit's Equations

Any integral of this equation involving p and q or both along with the given pde will give the values of p and q . Then by using $dz = p dx + q dy$ we can find the solution.

Problems:

Using Charpit's method solve the following equations.

i) $z = px + qy + pq$

iv) $z^2 = pqxy$

ii) $(p+q^2)y = qz$

v) $2z + p^2 + qy + 2y^2 = 0$

iii) $2(z + px + qy) = y^2 p$

vi) $2xz + pq - 2xyq - x^2p = 0$

Solutions

i) Here $F = px + qy + pq - z$

Charpit's equation is

$$\frac{dp}{F_x + F_z p} = \frac{dq}{F_y + F_z q} = \frac{dz}{-F_p - q F_q} = \frac{dx}{-F_p} = \frac{dy}{-F_q}$$

$$F_x = p, F_y = q, F_z = -1, F_p = x + q, F_q = y + p$$

$$\frac{dp}{p-p} = \frac{dq}{q-q} = \frac{dz}{-p(x+q) - q(y+p)} = \frac{dx}{-(x+q)} = \frac{dy}{-(y+p)}$$

$$\text{Consider } \frac{dp}{0} = \frac{dq}{0} \Rightarrow p = a, q = b$$

$$\therefore \text{complete integral is } z = \underline{ax + by + ab}$$

$$ii) \quad F = (p^2 + q^2)y - q^2 \quad \text{--- (1)}$$

$$f_x = 0, \quad f_y = p^2 + q^2, \quad f_z = -q, \quad f_p = 2py, \quad f_q = 2qy - 2$$

$$\frac{dp}{f_x + f_z p} = \frac{dq}{f_y + f_z q} = \frac{dz}{-pf_p - qf_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

$$\frac{dp}{-qp} = \frac{dq}{p^2 + q^2 - q^2} = \frac{dz}{-2p^2y - 2q^2y + 2} = \frac{dx}{-2py} = \frac{dy}{-2qy + 2}$$

$$\text{Consider } \frac{dp}{-qp} = \frac{dq}{p^2} \Rightarrow p dp = -q dq$$

$$\Rightarrow p dp + q dq = 0 \quad \text{Integrating we get } p^2 + q^2 = a^2 \quad \text{--- (2)}$$

$$\therefore (1) \Rightarrow a^2 y - q^2 = 0 \Rightarrow q = \frac{a^2 y}{2}$$

$$\therefore (2) \Rightarrow p^2 = a^2 - \left(\frac{a^2 y}{2}\right)^2 = \frac{a^2 z^2 - a^4 y^2}{2^2}$$

$$\therefore p = \frac{a\sqrt{z^2 - a^2 y^2}}{2}$$

$$\text{Now } dz = p dx + q dy$$

$$dz = \frac{a\sqrt{z^2 - a^2 y^2}}{2} dx + \frac{a^2 y}{2} dy$$

$$2 dz - a^2 y dy = a\sqrt{z^2 - a^2 y^2} dx$$

$$\Rightarrow \frac{2 dz - a^2 y dy}{\sqrt{z^2 - a^2 y^2}} = a dx$$

$$\text{Integrating, } \sqrt{z^2 - a^2 y^2} = ax + b$$

$$\Rightarrow \underline{z^2 - a^2 y^2 = (ax + b)^2}, \text{ which is the required complete int.}$$

$$iii) \quad F = z + px + qy - yp^2 = 0 \quad \text{--- (1)}$$

$$f_x = 2p, \quad f_y = 2q - p^2, \quad f_z = 1, \quad f_p = x - 2yp, \quad f_q = 2y$$

$$\frac{dp}{f_x - f_z p} = \frac{dq}{f_y - f_z q} = \frac{dz}{f_z} = \frac{dx}{f_p} = \frac{dy}{f_q}$$

$$\frac{dp}{2p + 2p} = \frac{dq}{2q - p^2 + 2q} = \frac{dz}{-2xp + 2yp^2 - 2qy} = \frac{dx}{-2x - 2yp} = \frac{dy}{-2y}$$

$$\text{Consider } \frac{dp}{2p} = \frac{dy}{-2y} \Rightarrow \frac{dp}{p} = -\frac{dy}{y}$$

$$\text{Integrating, } \log p = -2 \log y + \log a$$

$$\log p = \log(y^{-2} a) \Rightarrow p = \frac{a}{y^2}$$

$$\textcircled{1} \Rightarrow z + \frac{a}{y^2}x + qy = y \cdot \frac{a^2}{y^4}$$

$$2qy = \frac{a^2}{y^3} - 2z - \frac{2a}{y^2}x \Rightarrow q = \frac{a^2}{2y^4} - \frac{z}{y} - \frac{a}{y^3}x$$

$$\text{Now } dz = p dx + q dy$$

$$dz = \frac{a}{y^2} dx + \left(\frac{a^2}{2y^4} - \frac{z}{y} - \frac{ax}{y^3} \right) dy$$

$$\frac{y dz + z dy}{y} = \frac{ay dx - ax dy}{y^3} + \frac{a^2}{2y^4} dy$$

$$y dz + z dy = \frac{a(y dx - x dy)}{y^2} + \frac{a^2}{2y^3} dy$$

$$\text{Integrating, } yz = a\left(\frac{x}{y}\right) - \frac{a^2}{4y^2} + b$$

$$f = z - pqxy = 0 \quad \text{--- (1)}$$

$$f_x = -pqy, \quad f_y = -pqx, \quad f_z = z, \quad f_p = -qxy, \quad f_q = -pxy.$$

$$\frac{dp}{f_x + f_z p} = \frac{dq}{f_y + f_z q} = \frac{dz}{f_z} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

$$\frac{dp}{-pqy + z p} = \frac{dq}{-pqx + z q} = \frac{dz}{pqxy + pqxy} = \frac{dx}{qxy} = \frac{dy}{pxy}$$

Using the multipliers $(x, y, 0, p, q)$ we ve

$$\frac{x dp}{-pqxy + z p x} = \frac{y dq}{-pqxy + z q y} = \frac{p dx}{pqxy} = \frac{q dy}{pqxy}$$

$$\frac{x dp + y dq}{z x z p} = \frac{y dq + q dy}{z z q y}$$

$$\frac{x dp + p dx}{x p} = \frac{y dq + q dy}{y q}$$

Integrating, $\log(xp) = \log(yq) + \log a^2$
 $xp = yqa^2 \Rightarrow q = \frac{xp}{ya^2}$

$$\textcircled{1} \Rightarrow z^2 = p \left(\frac{xp}{ya^2} \right) xy \Rightarrow p^2 = \frac{a^2 z^2}{x^2} \Rightarrow p = \frac{az}{x}$$

$$\therefore q = \frac{x}{ya^2} \cdot \frac{az}{x} = \frac{z}{ay}$$

Now $dz = p dx + q dy$

$$\Rightarrow dz = \frac{az}{x} dx + \frac{z}{ay} dy$$

$$\frac{dz}{z} = \frac{a}{x} dx + \frac{dy}{y}$$

Integrating $\log z = a \log x + \frac{1}{a} \log y + \log b$
 $\Rightarrow \underline{z = b x^a y^{1/a}}$

v)

$$F = z^2 + p^2 + qy + 2y^2 = 0 \quad \text{--- (1)}$$

$$f_x = 0, \quad f_y = q + 4y, \quad f_z = 2z, \quad f_p = 2p, \quad f_q = y$$

$$\frac{dp}{f_x + f_z p} = \frac{dq}{f_y + f_z q} = \frac{dz}{p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

$$\frac{dp}{2p} = \frac{dq}{q + 4y + 2q} = \frac{dz}{-2p^2 - qy} = \frac{dx}{-2p} = \frac{dy}{-y}$$

Consider $\frac{dp}{2p} = \frac{dx}{-2p} \Rightarrow dp = -dx$

On integrating, $p = -x + a$

$$\therefore \textcircled{1} \Rightarrow z^2 + (a-x)^2 + qy + 2y^2 = 0$$

$$qy = -2y^2 - z^2 - (a-x)^2$$

$$q = \frac{1}{y} [-2y^2 - z^2 - (a-x)^2]$$

Now $dz = p dx + q dy$

$$dz = (a-x)dx + \frac{1}{y} [-2y^2 - z^2 - (a-x)^2] dy$$

Multiply both sides by $2y^2$

$$2y^2 dz = 2y^2(a-x)dx + 2y(-2y^2 - z^2 - (a-x)^2)dy$$

$$2y^2 dz + 4y^2 dy = 2y^2(a-x)dx - 2y(a-x)^2 dy - 4y^3 dy$$

$$d(2y^2 z) = d(-y^2(a-x)^2) - 4y^3 dy$$

On integrating

$$ay^2z = -y^2(a-x)^2 - y^2 + b.$$

vi) $F = 2xz + pq - 2xyq - x^2p \quad \text{--- (1)}$

$$f_x = 2z - 2yq, \quad f_y = -2xq, \quad f_z = 2x, \quad f_p = q - x^2, \quad f_q = p - 2xy.$$

$$\frac{dp}{f_x + f_z p} = \frac{dq}{f_y + f_z q} = \frac{dz}{-pf - qf_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

$$\frac{dp}{2z - 2yq + 2xz} = \frac{dq}{-2xq + 2xq} = \frac{dz}{-pq + px^2 - pq + 2xyq} = \frac{dx}{-q + x^2} = \frac{dy}{p + 2xy}$$

$$\frac{dq}{0} \Rightarrow q = a$$

$$\textcircled{1} \Rightarrow 2xz + ap - 2xya - x^2p = 0$$

$$p(a - x^2) = 2xya - 2xz \Rightarrow p = \frac{2x(ya - z)}{a - x^2}$$

Now $dz = p dx + q dy$

$$\Rightarrow dz = \frac{2xya - 2xz}{a - x^2} dx + a dy$$

$$= \frac{2x(z - ay)}{x^2 - a} dx + a dy.$$

$$\frac{dz - a dy}{z - ay} = \frac{2x}{x^2 - a} dx$$

On integrating, $\log(z - ay) = \log(x^2 - a) + \log b$

$$\text{or } \underline{\underline{z - ay = b(x^2 - a)}}$$

Method of Separation of Variables.

In this method, we assume that the dependent variable is the product of two functions, each of which involves only one of the independent variables. So two ordinary differential equations are formed.

Problems

Using the method of separation of variables solve the following equations.

i) $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$, given $u(x, 0) = 6e^{-3x}$.

ii) $\frac{\partial^2 u}{\partial x^2} - 2 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$.

iii) $\frac{\partial^2 z}{\partial x^2} = \frac{\partial z}{\partial y} + 2z$

iv) $\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}$, $u(0, y) = 8e^{-3y}$

v) $x \frac{\partial u}{\partial x} - 2y \frac{\partial u}{\partial y} = 0$.

vi) $u_x + u_y = 0$

Solutions.

i) $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$ — (1)

Here $u = f(x, t)$

Let $u = x(x) \cdot T(t)$ be a trial solution of (1)

Now $\frac{\partial u}{\partial x} = x' T$ and $\frac{\partial u}{\partial t} = x T'$

$\therefore (1) \Rightarrow x' T = 2 x T' + x T$
 $x' T = x(2 T' + T)$

$$\frac{x'}{x} = \frac{2T'+1}{T} \quad \text{--- (2)}$$

LHS of (2) is a function of x alone and RHS is a function of t alone. This can only be done if each side is equal to the same constant, say k , which is called separation constant. i.e. $\frac{x'}{x} = \frac{2T'+1}{T} = k$

$\frac{x'}{x} = k$ $\frac{(dx)}{dx} = k$ $\frac{dx}{x} = k dx$ $\Rightarrow \log x = kx + \log c_1$ $x = e^{kx + \log c_1}$ $x = c_1 e^{kx}$	$\frac{2T'+1}{T} = k \Rightarrow \frac{2T'}{T} = k-1$ $\Rightarrow \frac{T'}{T} = \frac{k-1}{2}$ $\Rightarrow \frac{dT}{T} = \left(\frac{k-1}{2}\right) dt$ $\Rightarrow \log T = \left(\frac{k-1}{2}\right) t + \log c_2$ $T = e^{\frac{k-1}{2}t + \log c_2}$ $= c_2 e^{\frac{k-1}{2}t}$
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$\therefore$ general solution is given by

$$u(x,t) = c_1 c_2 e^{kx} \cdot e^{(k-1)t/2} \quad \text{--- (3)}$$

Given $u(x,0) = 6e^{-3x}$

$$\textcircled{3} \Rightarrow u(x,0) = c_1 c_2 e^{kx} \quad \text{ie} \quad 6e^{-3x} = c_1 c_2 e^{kx}$$

$$\therefore c_1 c_2 = 6 \quad \text{and} \quad k = -3$$

$$\therefore u = 6e^{-3x} \cdot e^{-2t} = \underline{\underline{6e^{-(3x+2t)}}}$$

ii) $\frac{\partial^2 u}{\partial x^2} - 2 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$ — (1)

$u = f(x, y)$

Let $u = x(x) \cdot Y(y)$ be a trial solution of (1)

$\frac{\partial u}{\partial x} = x'Y \quad \frac{\partial^2 u}{\partial x^2} = x''Y \quad \frac{\partial u}{\partial y} = xY'$

(1) $\Rightarrow x''Y - 2x'Y + xY' = 0 \Rightarrow (x'' - 2x')Y = -xY'$

$\frac{x'' - 2x'}{x} = -\frac{Y'}{Y} = k$

$\frac{x'' - 2x'}{x} = k$

$\Rightarrow x'' - 2x' - kx = 0$

$\Rightarrow (D^2 - 2D - k)x = 0$ At $m^2 - 2m - k = 0$

$\Rightarrow m = 1 \pm \sqrt{1+k}$

$\therefore x = C_1 e^{(1+\sqrt{1+k})x} + C_2 e^{(1-\sqrt{1+k})x}$

$-\frac{Y'}{Y} = k \Rightarrow \frac{dY}{dY} = -kY \Rightarrow \frac{dY}{Y} = -k dY$

$\Rightarrow \log Y = -kY + \log C_3 \Rightarrow Y = C_3 e^{-kY}$

$\therefore u(x, y) = \left[C_1 e^{(1+\sqrt{1+k})x} + C_2 e^{(1-\sqrt{1+k})x} \right] C_3 e^{-kY}$

iii) $\frac{\partial^2 z}{\partial x^2} = \frac{\partial z}{\partial y} + 2z$ — (1)

$z = f(x, y)$

Let $z = x(x) \cdot Y(y)$ be a trial solution of (1)

$\frac{\partial z}{\partial y} = xY' \quad \frac{\partial^2 z}{\partial x^2} = x''Y$

$$① \Rightarrow x''y = xy' + 2xy$$

$$(x'' - 2x)y = xy' \Rightarrow \frac{x'' - 2x}{x} = \frac{y'}{y} = k$$

$$\frac{x'' - 2x}{x} = k$$

$$\Rightarrow \frac{x''}{x} = k+2$$

$$\Rightarrow x'' - (k+2)x = 0$$

$$\Rightarrow [D^2 - (k+2)]x = 0$$

$$\Delta E: m^2 - (k+2) = 0$$

$$m = \pm \sqrt{k+2}$$

$$x = C_1 e^{\sqrt{k+2}x} + C_2 e^{-\sqrt{k+2}x}$$

$$\therefore z = \left(C_1 e^{\sqrt{k+2}x} + C_2 e^{-\sqrt{k+2}x} \right) C_3 e^{ky}$$

$$\frac{y'}{y} = k$$

$$\Rightarrow \frac{dy}{dy} = ky$$

$$\Rightarrow \frac{dy}{y} = k dy$$

$$\Rightarrow \log y = ky + \log C_3$$

$$y = C_3 e^{ky}$$

$$ii) \quad x \frac{\partial y}{\partial x} - 2y \frac{\partial y}{\partial y} = 0 \quad \text{--- (1)}$$

$u = x(x) y(y)$ be trial solution of (1)

$$\frac{\partial y}{\partial x} = x'y \quad \frac{\partial y}{\partial y} = xy'$$

$$① \Rightarrow x x'y - 2y \cdot xy' = 0 \Rightarrow \frac{xx'}{x} = 2y \frac{y'}{y} = k$$

$$x \frac{x'}{x} = k$$

$$\Rightarrow \frac{dx}{dx} = \frac{x}{x} k$$

$$\Rightarrow \frac{dx}{x} = k \cdot \frac{dx}{x}$$

$$\Rightarrow \log x = k \log x + \log e_1$$

$$x = C_1 e^{kx}$$

$$2y \frac{y'}{y} = k$$

$$\Rightarrow \frac{dy}{dy} = \frac{y k}{2y}$$

$$\Rightarrow \frac{dy}{y} = \frac{dy k}{2y}$$

$$\Rightarrow \log y = \frac{k}{2} \log y + \log e_2$$

$$y = C_2 y^{k/2}$$

$$\therefore u = C_1 C_2 x^{k/2} y^{k/2}$$

$$v) \quad \frac{\partial u}{\partial x} = 1 \frac{\partial u}{\partial y} \quad \text{--- (1)}$$

$u = x(y)$ be trial solution of (1)

$$\frac{\partial u}{\partial x} = x' y \quad \frac{\partial u}{\partial y} = x y'$$

$$(1) \Rightarrow x' y = 1 x y' \Rightarrow \frac{x'}{x} = 1 \frac{y'}{y} = k$$

$$\frac{x'}{x} = k$$

$$\frac{y'}{y} = \frac{k}{1}$$

$$\log x = kx + \log c_1$$

$$\log y = \frac{k}{1} y + \log c_2$$

$$x = c_1 e^{kx}$$

$$y = c_2 e^{\frac{k}{1} y}$$

$$\therefore u = \underline{\underline{c_1 c_2 e^{kx} e^{\frac{k}{1} y}}} \quad \text{--- (2)}$$

Given $u(0, y) = 8 e^{-3y}$

$$(2) \Rightarrow u(0, y) = c_1 c_2 e^{\frac{k}{1} y} \quad \text{is} \quad 8 e^{-3y} = c_1 c_2 e^{\frac{k}{1} y}$$

$$\therefore c_1 c_2 = 8, \quad \frac{k}{1} = -3 \Rightarrow k = -12$$

$$\therefore u = 8 e^{-12x} e^{-3y} = \underline{\underline{8 e^{-3(4x+y)}}}$$

vi)

$$u_x + u_y = 0 \Rightarrow \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0 \quad \text{--- (1)}$$

$u = xy$ be trial solution of (1)

$$\frac{\partial u}{\partial x} = x' y \quad \frac{\partial u}{\partial y} = x y'$$

$$(1) \Rightarrow x' y + x y' = 0 \Rightarrow \frac{x'}{x} = -\frac{y'}{y} = k$$

$$\frac{x'}{x} = k \Rightarrow \log x = kx + \log c_1 \Rightarrow x = c_1 e^{kx}$$

$$\frac{y'}{y} = -k \Rightarrow \log y = -ky + \log c_2 \Rightarrow y = c_2 e^{-ky}$$

Question Bank.

I Form pde from the following equations by eliminating arbitrary constants.

i) $z = ax + by + (a^2 + b^2)^2$ (ii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

iii) $z = (x^2 + a)(y^2 + b)$ (iv) $(x-a)^2 + (y-b)^2 + z^2 = 1$

v) $z = ax^2 + bxy + cy^2$ (vi) $z = axy + (a^2 + b^2)y$

II Obtain pde by eliminating the arbitrary functions from the following.

i) $z = f\left(\frac{xy}{z}\right)$ (ii) $xyz = \phi(x+y+z)$

iii) $x^2 + y^2 + z^2 = f(xy)$ (iv) $z = f(x) + e^y g(x)$

v) $\phi(x^2 - y^2, x^2 - z^2) = 0$ (vi) $f(x+y+z, x^2 + 2yz) = 0$

III Solve the following equations

i) $\frac{\partial^2 z}{\partial x \partial y} = \cos(ax + by)$ (ii) $\frac{\partial^2 z}{\partial y^2} = \cos(xy)$

iii) $\frac{\partial^2 z}{\partial x^2} + a^2 z = 0$, given that when $x=0$, $z=e^y$ and $\frac{\partial z}{\partial x} = a$

iv) $\frac{\partial^2 z}{\partial y^2} = z$, given that when $y=0$, $z=e^x$ and $\frac{\partial z}{\partial y} = e^{-x}$

IV Solve the following

i) $(y^2 + z^2)p - xyq + xz = 0$ (ii) $\frac{y-z}{yz}p + \frac{z-x}{zx}q = \frac{x-y}{xy}$

iii) $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$ (iv) $xp + yq = z$

v) $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$ (vi) $\frac{p}{x^2} + \frac{q}{y^2} = z$

V Solve the following by Charpit's method

i) $px + qy = pq$ (ii) $y^2 p^2 = q$ (iii) $z = p^2 x + q^2 y$

iv) $x(px + qy) = z(1 - q^2)$